Pontificia Universidad Católica de Chile
IA Lab

Applications: Summary Generation

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Intro

Motivation

- Summarizing is ubiquitous. Movie trailers and plots, book plots, elevator pitches, headlines, paper abstracts, etc.
- Help us extract the most important points and focus on them discarding noise or details.
- Huge time saver in various situations.
- Manual text summarization is a time expensive and generally laborious task.

Today's Class

- We are going to talk about the automatic summarization task.
- We are going to focus exclusively on text.
- Different flavours.
- Different ways to tackle the problem.
- How to measure performance.

Table of Contents

Task

Given an input text x, we want to produce a summary y which is shorter and includes the most important information present in x.

Abilities

To be able to summarize we need to:

- Identify the most important ideas.
- Ignore irrelevant information.
- Integrate these ideas in a meaningful way.

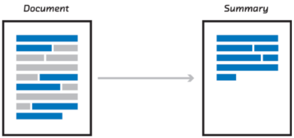
Why hard?

Some of these ideas can be contextually important, or unimportant.

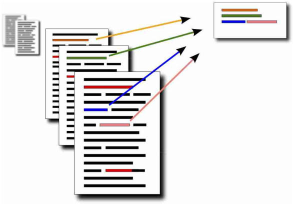
Commonsense can play a big role.

The task can by divided into:

- **Single-Document summarization.** Generate a summary from a single document x.



- **Multi-document summarization.** Generate a summary from a multiple documents $x_1, x_2, ..., x_n$.



How can we approach this problem?

Roughly two approaches

- **Extractive summarization.** Summarizing by detecting the most important parts of the text. Similar to text highlighting.
- **Abstractive summarization.** Summarize by paraphrasing the text.

Comparison

Extractive summarization.

- Easier.
- Restrictive.
- No room for paraphrasing.

Abstractive summarization.

- Harder.
- Flexible.
- More human.

Data

Some of the most commonly used datsets [1]:

- **CNN/DailyMail.** 312.000 news articles with multi-sentences summaries.
- **Gigaword.** Headline generation on a corpus of 4 million of articles from newswire.
- **LCSTS (Large Scale Chinese Short Text Summarization Dataset).** 2 million chinese short texts from the microblogging website Sina Weibo with short summaries given by the author.
- **X-Sum.** Extreme summarization of news into a short one-sentence summary composed of 225.000 examples.
- **Wikihow.** Based on the website Wikihow, sumarize 200.000 procedure texts into their headline.

[1] At this repo you can find over 22 different datasets.

Summarization is a Heterogeneous Task

- **CNN/DailyMail.**

  article $\longrightarrow$ multi-sentences summaries

- **Gigaword.**

  first sentences of article $\longrightarrow$ headline

- **LCSTS.**

  paragraph $\longrightarrow$ sentence

- **X-Sum.**

  article $\longrightarrow$ one-sentence summary

- **Wikihow.**

  article $\longrightarrow$ summary sentences

CNN/DailyMail

Document (truncated):

Lagos , nigeria (cnn) a day after winning nigeria 's presidency , muhammadu buhari told cnn 's christiane amanpour that he plans to aggressively fight corruption that has long plagued nigeria and go after the root of the nation 's unrest . buhari said he 'll “ rapidly give attention ” to curbing violence in the northeast part of nigeria , where the terrorist group boko haram operates . by cooperating with neighboring nations chad , cameroon and niger , he said his administration is confident it will be able to thwart criminals and others contributing to nigeria 's instability . for the first time in nigeria 's history , the opposition defeated the ruling party in democratic elections . buhari defeated incumbent goodluck jonathan by about 2 million votes , according to nigeria 's independent national electoral commission . the win comes after a long history of military rule , coups and botched attempts at democracy in africa 's most populous nation .

Reference summary:

Muhammadu buhari tells cnn 's christiane amanpour that he will fight corruption in nigeria . nigeria is the most populous country in africa and is grappling with violent boko haram extremists . nigeria is also africa 's biggest economy , but up to 70 % of nigerians live on less than a dollar a day .

Document:
The suicide bomb attacks in saudi arabia were a cowardly and disgraceful terrorist atrocity , " prime minister tony blair said wednesday .

Reference summary:
Two britons missing after saudi suicide blasts.

Document:
Zairean rebels , led by laurent-desire kabila , on saturday rejected calls by the united nations for a ceasefire , saying it could only be called after talks with kinshasa .

Reference summary:
Zairean rebels reject un call for ceasefire.

How to build a dataset?

- Web scraping: Take advantage of large amounts on data in te web
- Look for domains in which summarizing naturally occurs (news, social networks, etc.)
- The closer the domain to yours, the better.

Possible summarization sources

- News articles and their headlines.
- Previews (movies).
- Digest (TV guides).
- Wikipedia (first section).
- Academic papers (abstract).

Extractive Model

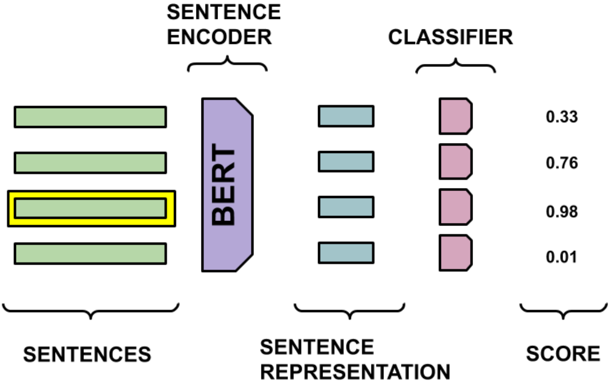
Task framing

The task of extractive summarization is to select the most relevant sentences from a given text in order to use them as the target summary.

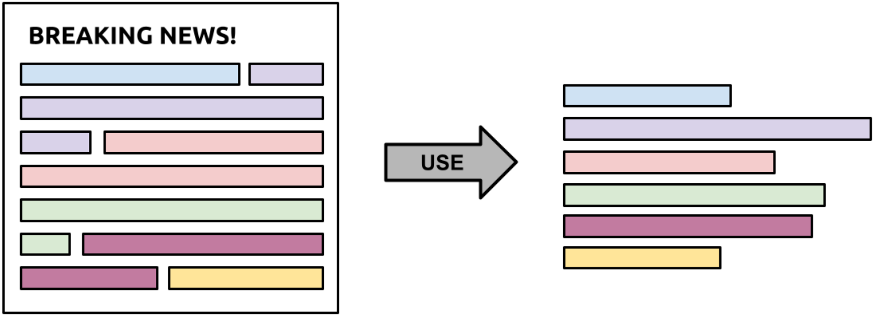
For this we need to:

1. Find a way to represent our sentences or fragments.
2. Score each sentence or fragment.
3. Select a sub-set of sentences/fragments to create the summary.

Extractive Model

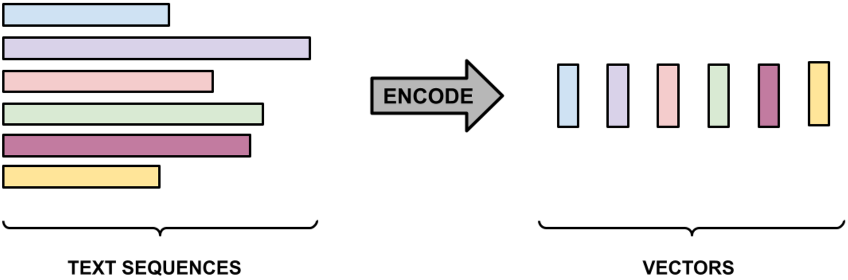


Extractive Model



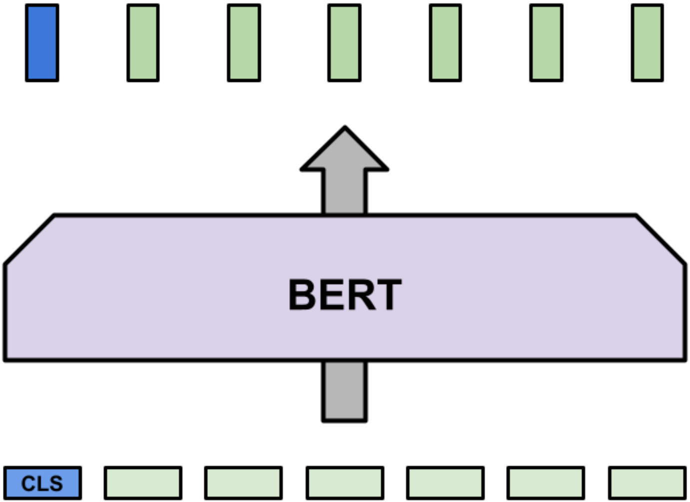
Extractive Model

1. Find a way to represent our sentences or fragments.



Extractive Model

Encode using BERT.



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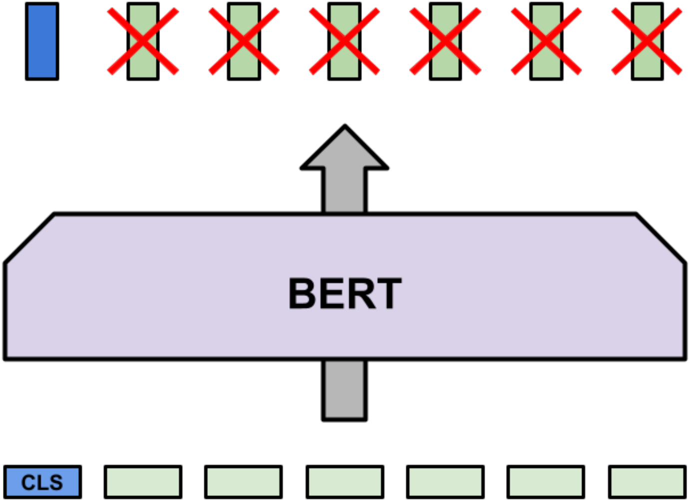

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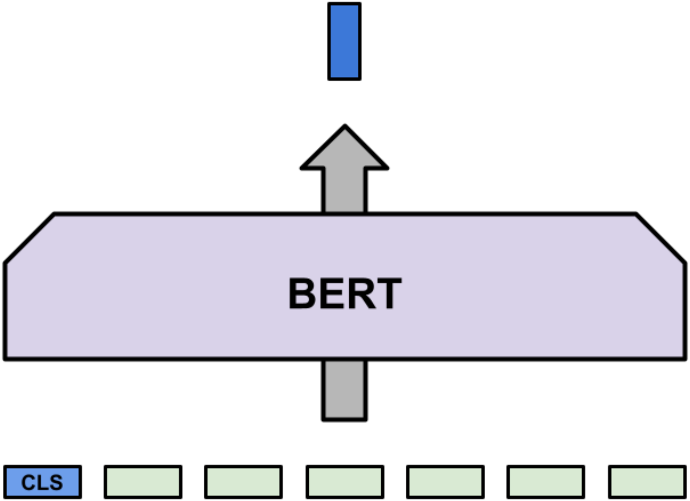
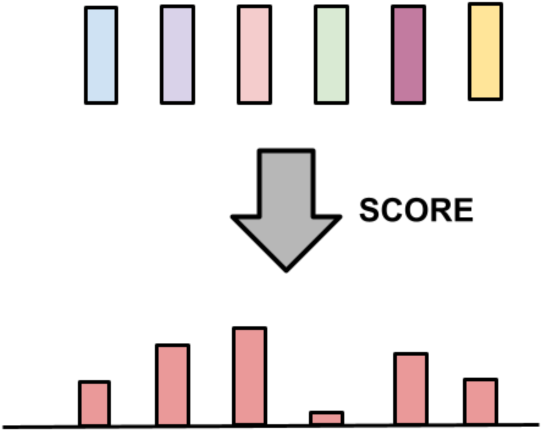
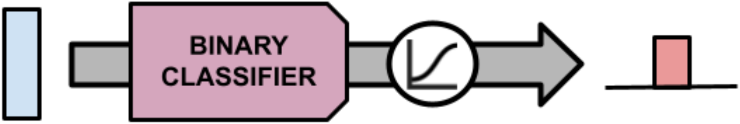


Extractive Model

2. Score each sentence or fragment.



Learn a binary classifier for scoring.



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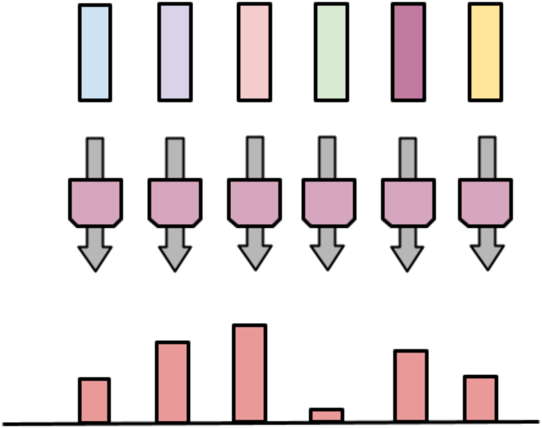
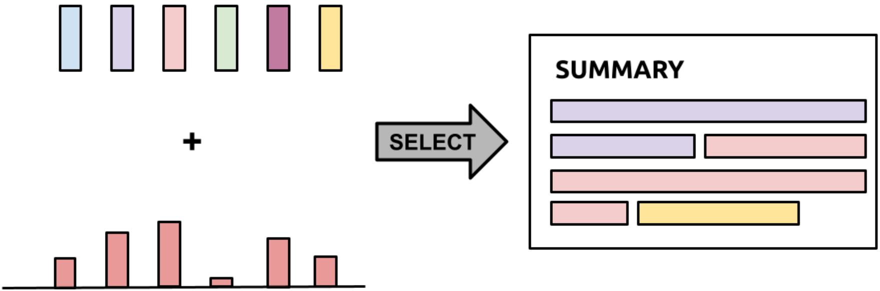


What ground truth to use?

What ground truth to use?

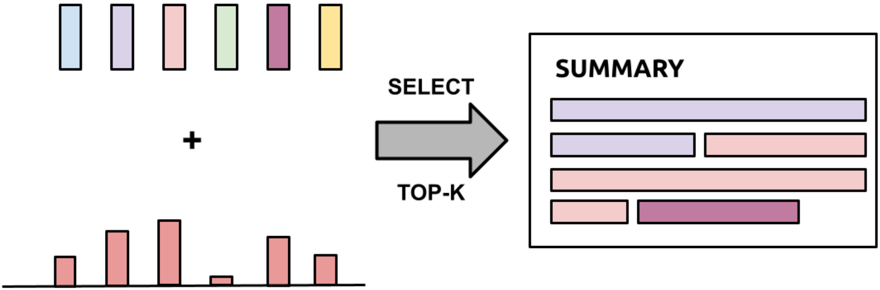
Choose the subset of sentences that maximize a metric (ROUGE) as ground truth sentences.

3. Select a sub-set of sentences/fragments to create the summary.



Extractive Model

Select the top-k highest scores to form the summary.

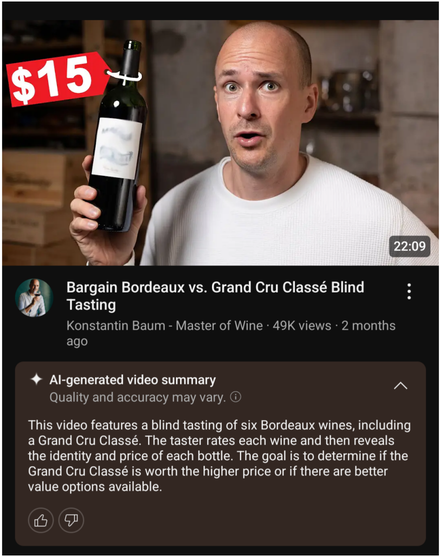


Extractive Summarization Use Cases

- Legal Document Summarization
- Summarizing Court Proceedings
- Financial Reports and Market Summaries
- Healthcare Records and Medical Summaries
- Meeting Minutes and Transcripts

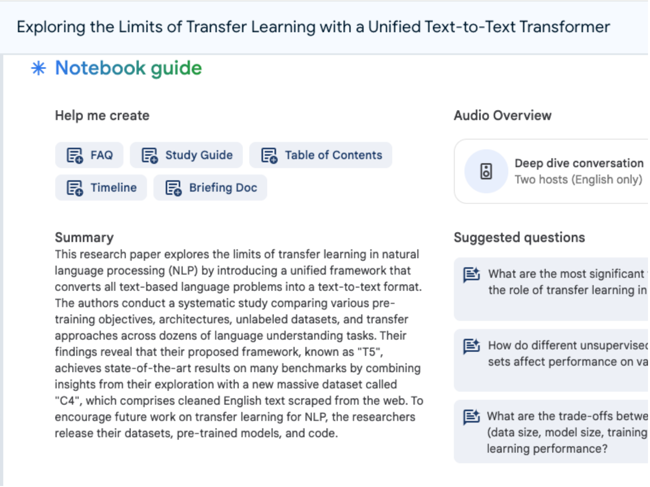
Abstractive Model

Abstractive Summarization Applications



Abstractive Summarization Applications

NotebookLM



T5 Model

Journal of Machine Learning Research 21 (2020) 1-67 Submitted 1/20; Revised 6/20; Published 6/20

Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer

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"**T**ext-**to**-**T**ext **T**ransfer **T**ransformer"

Idea: We can solve any NLP task using the same model.

Based on an Encoder-Decoder Transformer model.

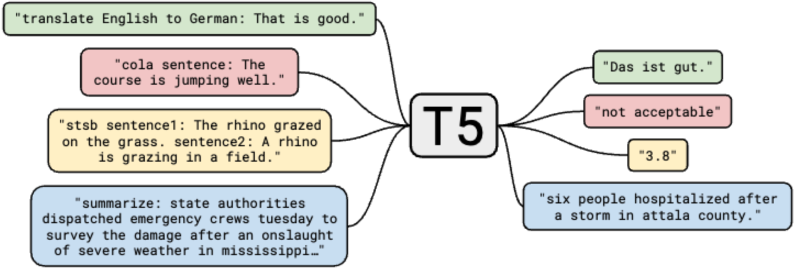


Figure from "Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer" paper.

A multi-task mixture of unsupervised and supervised training.

Unsupervised Training:

- Span-corruption objective: randomly removing 15% of the tokens.
- Colossal Clean Crawled Corpus (C4)[2]

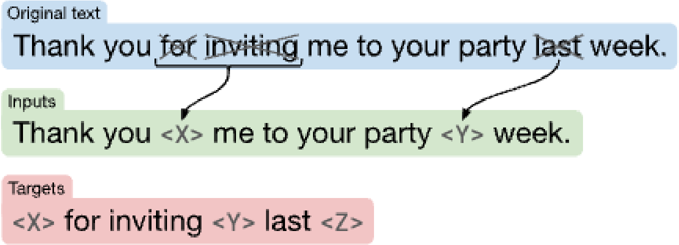


Figure from "Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer" paper.

[2] https://www.tensorflow.org/datasets/catalog/c4

T5 Model Training

Supervised Training:

- Multi-Task training.
- Many datasets such as:
  - **CNN/Daily Mail**
  - GLUE
  - SuperGLUE
  - SQuAD
  - WMT English to German, French, and Romanian translation

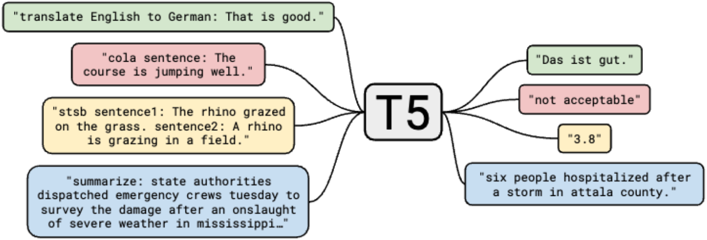


Figure from "Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer" paper.

Some Results (only for summarization):

Model	CNN/DM ROUGE-1	CNN/DM ROUGE-2	CNN/DM ROUGE-L
Previous best	43.47[g]	20.30[g]	40.63[g]
T5-Small	41.12	19.56	38.35
T5-Base	42.05	20.34	39.40
T5-Large	42.50	20.68	39.75
T5-3B	42.72	21.02	39.94
T5-11B	**43.52**	**21.55**	**40.69**

New T5 models have been trained on multilingual datasets:

- MT5[3]
- UMT5[4]

Both use multilingual versions of C4 to improve performance in non-English languages.

However, both are only based on an unsupervised objective, so they need finetuning.

[3] https://huggingface.co/docs/transformers/model_doc/mt5
[4] https://huggingface.co/docs/transformers/model_doc/umt5

Text Generation

Decoding

Models are trained to generate a probability distribution based on the text to summarize and the previous generated tokens:

$$P(y_t|x, y_{<t}) = f(x, y_{<t})$$

We need an algorithm (g) to use this distribution to generate new tokens:

$$\hat{y}_t = g(P(y_t|x, y_{<t}))$$

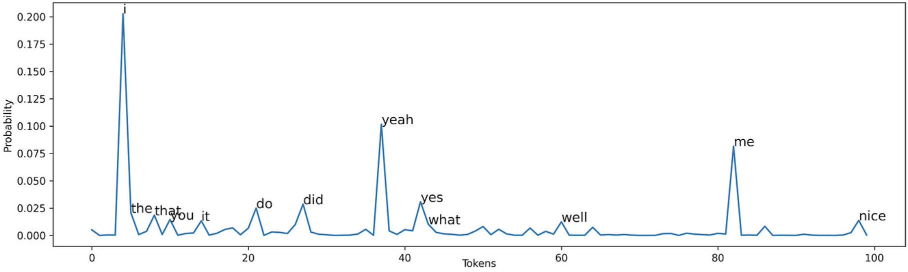
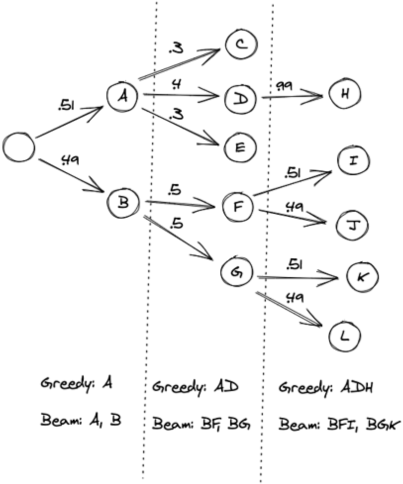


Figure from: https://towardsdatascience.com/decoding-strategies-that-you-need-to-know-for-response-generation-ba95ee0faadc

Greedy Decoding: Select always the token with the highest probability.

$$\hat{y}_t = \arg\max_{w} P(y_t = w | x, y_{<t})$$

Beam Search: Keep k candidates (or beams) in every decoding step



Decoding: Isssues

Context

In a shocking finding, scientist discovered a herd of unicorns living in a remote, previously unexplored valley, in the Andes Mountains. Even more surprising to the researchers was the fact that the unicorns spoke perfect English.

Beam Search

"The study, published in the Proceedings of the National Academy of Sciences of the United States of America (PNAS), was conducted by researchers from the Universidad Nacional Autónoma de México (UNAM) and the Universidad Nacional Autónoma de México (UNAM/Universidad Nacional Autónoma de México/Universidad Nacional Autónoma de México/Universidad Nacional Autónoma de México/Universidad Nacional Autónoma de ..."

(Holtzman et al. ICLR 2020)

Introduce randomness to decoding.

Top-p (nucleus) Sampling: Sample from the tokens for which their cumulative probability less than p.

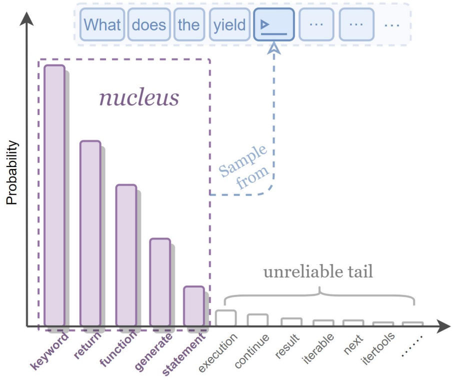
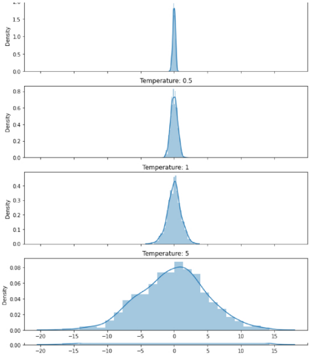


Figure from: https://ar5iv.labs.arxiv.org/html/2208.11523



What if we want to add more randomness, i.e. increase variability?

We can increase the temperature!

$$P(y_t = w) = \frac{\exp(u_w/t)}{\sum_j \exp(u_j/t)}$$

Metrics

ROUGE

Recall-Oriented Understudy for Gisting Evaluation

The main metric to test summarization.

Not as good as human evaluation, but more convenient.

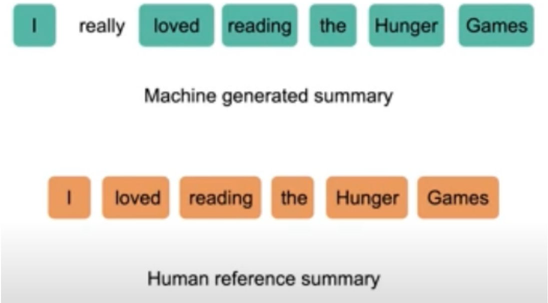
Usually reported separately for each n-gram:

ROUGE-1, ROUGE-2, ROUGE-L, etc...

ROUGE-L measures the overlap for the longest common subsequence.

In Python: `pip install rouge-score`.

Rouge-1



$$\text{Rouge-1 Recall} = \frac{\text{Num correct words}}{\text{Num words in reference}} = \frac{6}{6}$$

$$\text{Rouge-1 Precision} = \frac{\text{Num correct words}}{\text{Num words in summary}} = \frac{6}{7}$$

$$\text{Rouge-1 F1} = 2\left(\frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}\right) = \frac{12}{13}$$

Slide from: https://www.youtube.com/watch?v=TMshhnrEXlg

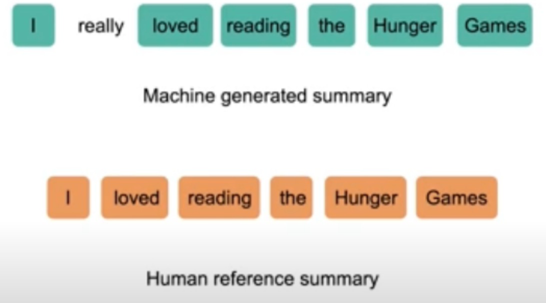
Rouge-2

$$\text{Rouge-2 Recall} = \frac{4}{5}$$

$$\text{Rouge-2 Precision} = \frac{4}{6}$$

Slide from: https://www.youtube.com/watch?v=TMshhnrEXlg

Rouge-L



$$\text{Rouge-L Recall} = \frac{LCS(s,r)}{\text{Num words in reference}} = \frac{6}{6}$$

$$\text{Rouge-L Precision} = \frac{LCS(s,r)}{\text{Num words in summary}} = \frac{6}{7}$$

Slide from: https://www.youtube.com/watch?v=TMshhnrEXlg

Final words

Great resources

- Article on how to generate text using Huggingface Transformers.
- Article on how to do summarization with Huggingface Transformers.
- Article by AssemblyAI comparing summarization APIs and more.
- Video of Stanford class on text generation.
- Video explaining ROUGE score.
- Huggingface summarization models.
- `https://github.com/mathsyouth/awesome-text-summarization`

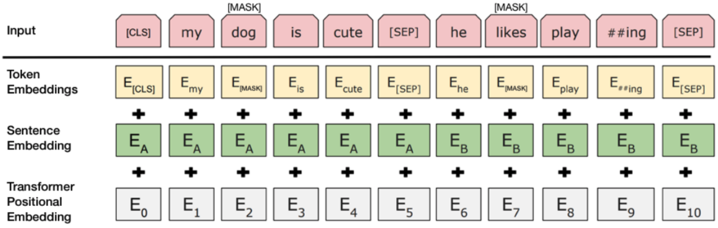
Questions?

Appendix

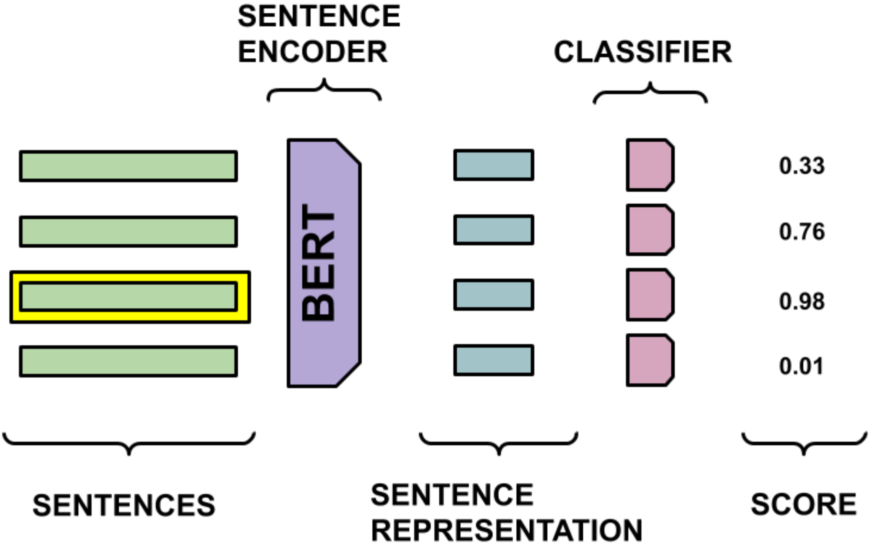
Fine-tune BERT for Extractive Summarization by Yang Liu.

Use a pretrained BERT model to represent each sentence.

Fine-tune it to score each sentence.



Extractive Model: Summary



We need to provide a score to our model in order to learn.

This work: The combination of text sentences which maximize the rouge score.

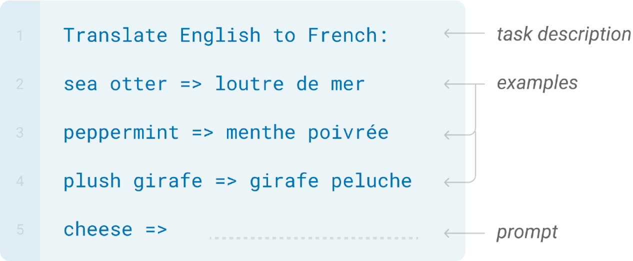
Other alternatives

- Best case: Having a ground truth.
- Frequency based scores, i.e. word probabilities, tf-idf.
- Information theoretic based, i.e. KL-Divergence.

Prompt Engineering

Finding the correct way to describe a task in order for a LLM can solve it correctly.

Prepending solved examples of the task before our example. **In-context Learning!**



Be careful: Language models are not trained to assist users.

Choose instruction tuned LLMs. Example: `gpt-3.5-turbo-instruct`

PROMPT *Explain the moon landing to a 6 year old in a few sentences.*

COMPLETION GPT-3

```
Explain the theory of gravity to a 6 year old.

Explain the theory of relativity to a 6 year old in a few sentences.

Explain the big bang theory to a 6 year old.

Explain evolution to a 6 year old.
```

Slide inspired by: Stanford CS224N | 2023 | Lecture 10 - Prompting, Reinforcement Learning from Human Feedback

Provide an instruction.

```
prompt = f"""
Please provide a summary of the following text:

TEXT:

{text}
"""
```

Provide an instruction.

```
prompt = f"""

{text}

Explain the above in one sentence:
"""
```

Usually helps to be descriptive.

```
prompt = f"""
Please provide a summary of the following text.  Please
provide your output in a manner that a 5 year old would
understand:

TEXT:

{text}
"""
```

Usually helps to be descriptive.

```
prompt = f"""
Write a concise summary of the following:

{text}

CONCISE SUMMARY:
"""
```

Usually helps to be descriptive.

```
prompt = f"""
Write a concise summary of the following text delimited by
triple backquotes.  Return your response in bullet points
which covers the key points of the text.

{text}

BULLET POINT SUMMARY:
"""
```